

A 15-Qubit Decoherence-Reversal Protocol for Benchmarking Mid-Circuit Measurements and Environmental Imprinting

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Abstract As quantum processors scale toward the regime of utility, validating hardware performance increasingly relies on approximate classical simulations, such as tensor networks, which introduce truncation errors. This paper introduces a 15-qubit decoherence-reversal protocol designed to isolate physical hardware noise from algorithmic artifacts. The protocol employs a forward scrambling circuit, exploratory mid-circuit measurement for environmental tracking, and a Loschmidt-echo reversal. By operating within a state-vector tractable regime, this architecture provides a clean ideal-model baseline ($F_{\max} \approx 0.9889$) against which physical recovery can be directly compared. We demonstrate the diagnostic sensitivity of the protocol via a 400-shot production run on the Azure Quantum `quantinuum.sim.h2-1e` emulator. Calibrating the local surrogate to published H2-1 typical specifications predicted $F_{\text{expected}} = 0.8353 \pm 0.0054$, while the emulator returned a mitigated $F_{\text{recovered}} = 0.7836 \pm 0.0212$. The protocol successfully flagged the statistically significant shortfall as “Anomalous Decoherence.” In the context of an emulator backend, this result represents a successful detection of model mismatch, demonstrating the protocol’s sensitivity to unmodeled noise structures — consistent with unmodeled full-stack compilation/routing overhead or other emulator-side effects.

1 Introduction

The validation of near-term quantum hardware requires rigorous benchmarking protocols that can cleanly separate fundamental information loss from ordinary experimental noise [8]. Benchmarking at scale necessitates approximate classical methods, introducing tensor-network truncation as a confounding factor when evaluating hardware fidelity [9].

To address this, we present a 15-qubit dynamic circuit protocol restricted to a regime that remains exactly simulable at the state-vector level. This protocol combines a unitary scrambling sequence with an exploratory mid-circuit measurement layer, followed by a time-reversal sequence.

1.1 A Tractable Exact-Baseline Architecture

While much current work emphasizes larger systems that require approximate classical methods, this protocol is restricted to a regime that remains exactly simulable. This removes tensor-network truncation as a confounding factor and provides a clean ideal-model baseline ($F_{\max}$) against which physical recovery can be compared.

1.2 A Sensitive Diagnostic Benchmark

The protocol is a controlled benchmarking tool for hardware supporting mid-circuit measurements. Because the exact baseline can be computed directly and compared with surrogate noise thresholds, the protocol provides a structured way to study recovery shortfalls. If a physical processor returns recovery below the modeled range, the result motivates investigation of unmodeled noise

sources such as crosstalk, leakage, or calibration drift [7].

1.3 Foundations and Reversibility

The framework of environment-induced decoherence, particularly Zurek’s theory of Quantum Darwinism [1, 10], posits that classicality is an illusion born of information proliferation into unmeasured degrees of freedom. While previous work has observed imprinting in photonic cluster-state systems [3], demonstrating the strict reversal of a “scrambled” state remains a challenge akin to catching and reversing a quantum jump [4]. The combination of reversible scrambling with mid-circuit Quantum Non-Demolition (QND) measurements serves as a foundations-motivated probe of information reversibility in programmable systems [6].

2 Theoretical Framework

2.1 Einselection and Environmental Entanglement

Consider a system S interacting with an artificial environment E . Initially, S is in state $|\psi_0\rangle_S$. Interaction creates entanglement [2]:

$$|\Psi(t_1)\rangle_{SE} = \sum_k \alpha_k |\phi_k\rangle_S \otimes |E_k\rangle_E \quad (1)$$

where α_k are amplitudes and $|\phi_k\rangle_S$ are pointer states. Tracing over the environment yields a reduced density matrix for S :

$$\rho_S = \text{Tr}_E(|\Psi(t_1)\rangle\langle\Psi(t_1)|) \approx \sum_k |\alpha_k|^2 |\phi_k\rangle_S \langle\phi_k| \quad (2)$$

Applying the inverse Hamiltonian $U^\dagger$ returns the joint system toward its initial state:

$$U^\dagger |\Psi(t_1)\rangle_{SE} = |\psi_0\rangle_S \otimes |0\rangle_E \quad (3)$$

2.2 Recovery Fidelity and Exact Simulation

The Loschmidt echo quantifies this reversibility [5]:

$$L(t) = |\langle \psi_0 | U^\dagger(t) U(t) | \psi_0 \rangle|^2 \quad (4)$$

We operationalize recovery fidelity via the target 11-qubit ground state population, explicitly tracing out the probe register:

$$F_{\text{recovered}} = \langle 0^{\otimes 11} | \text{Tr}_P(\rho_{\text{final}}) | 0^{\otimes 11} \rangle \quad (5)$$

3 Experimental Protocol

The protocol isolates a 15-qubit subsystem: a 6-qubit system S , a 5-qubit environment E , and 4 QND probes P optimized for current trapped-ion hardware [9].

Table 1: Experimental Phase Protocol Summary

Phase	Objective / Action
I. Calibration	Define surrogate profiles based on manufacturer specifications.
II. Exact Sim.	Calculate exact F_{max} via branch summation.
III. Forward	Apply scrambling circuit $U(t)$ and coupling.
IV. Tracking	Mid-circuit QND probe readout.
V. Execution	Submit compiled OpenQASM 2.0 to hardware scheduler.
VI. Analysis	Quantitative threshold classification.

3.1 Hardware Execution Note

Submission to the Quantinuum H2-1E emulator required deviations from standard pathways to preserve mid-circuit measurements. Standard Azure QIR transpilation was found to mishandle multi-shot mid-circuit evaluations. To bypass this, the circuit was compiled locally to OpenQASM 2.0 via `qasm2.dumps()` and submitted using the bare-metal `Job.from_input_data()` API with honeywell format tags. Furthermore, the classical registers were flattened into a single 15-bit register to ensure deterministic bitstring concatenation in the JSON payload.

4 Results

4.1 Surrogate Expectation

Surrogate modeling was calibrated to published Quantinuum H2-1 specifications ($e_1 = 3 \times 10^{-5}$, $e_2 = 1 \times 10^{-3}$, $e_{\text{spam}} = 1 \times 10^{-3}$) and pessimistic limits.

Table 2: Sensitivity Sweep (H2-1 Calibrated). Column order: F_{clean} (zero-noise baseline) is the ~ 0.99 column; F_{exp} (noisy expectation) is the lower column.

Scenario	$F_{\text{clean}} \pm \text{SE}$	$F_{\text{exp}} \pm \text{SE}$	Degr. (δ)
H2 Typical Profile	0.9884 ± 0.0014	0.8353 ± 0.0054	0.1531
H2 Pessimistic Limit	0.9880 ± 0.0013	0.6805 ± 0.0071	0.3075
Deep Scramble (H2)	0.9902 ± 0.0013	0.7831 ± 0.0064	0.2071
Strong QND Kick (H2)	0.9556 ± 0.0029	0.8058 ± 0.0055	0.1498

4.2 Hardware Emulation Validation

A 400-shot run on `quantinuum.sim.h2-1e` recovered the ground state with a raw 76.0% frequency.

After SPAM mitigation, $F_{\text{recovered}} = 0.7836 \pm 0.0212$. The 0.0517 gap between expectation and recovery exceeded the combined 2σ band (± 0.0438), triggering the “Anomalous Decoherence” classification.

5 Discussion

The classification of the emulator run as “Anomalous Decoherence” represents a successful demonstration of the protocol’s sensitivity to model mismatch. It is important to distinguish this finding from evidence of physical hardware decoherence. Since the execution occurred on a vendor emulator, the flag identifies a statistically significant gap between the idealized local noise assumptions and the full-stack emulation model.

The shortfall is consistent with unmodeled full-stack compilation/routing overhead or other emulator-side effects. For instance, mapping a scrambled 15-qubit state to specific trapped-ion routing zones may require dynamic SWAP gates that increase physical depth beyond logical predictions. By successfully detecting this subtle variation, the diagnostic pipeline provides evidence for its utility as a high-resolution tool for evaluating the hidden overhead of commercial quantum stacks.

References

- [1] W. H. Zurek, “Quantum Darwinism,” *Nature Physics* **5**, 181–188 (2009).
- [2] Č. Brukner, “On the quantum measurement problem,” in *Quantum [Un]Speakables II*, eds. R. Bertlmann and A. Zeilinger (Springer, Cham, 2017), pp. 95–117.
- [3] M. A. Ciampini, G. Pinna, P. Mataloni, and M. Paternostro, “Experimental signature of quantum Darwinism in photonic cluster states,” *Phys. Rev. A* **98**, 020101(R) (2018).
- [4] Z. K. Mineev *et al.*, “To catch and reverse a quantum jump mid-flight,” *Nature* **570**, 200–204 (2019).
- [5] X. Mi *et al.*, “Time-crystalline eigenstate order on a quantum processor,” *Nature* **601**, 531–536 (2022).
- [6] M. S. Blok *et al.*, “Quantum Information Scrambling on a Superconducting Qutrit Processor,” *Phys. Rev. X* **11**, 021010 (2021).

- [7] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, “Circuit quantum electrodynamics,” *Rev. Mod. Phys.* **93**, 025005 (2021).
- [8] F. Arute *et al.*, “Quantum supremacy using a programmable superconducting processor,” *Nature* **574**, 505–510 (2019).
- [9] Y. Kim *et al.*, “Evidence for the utility of quantum computing before fault tolerance,” *Nature* **618**, 500–505 (2023).
- [10] A. Touil, B. Yan, D. Girolami, S. Deffner, and W. H. Zurek, “Eavesdropping on the Decohering Environment: Quantum Darwinism, Amplification, and the Origin of Objective Classical Reality,” *Phys. Rev. Lett.* **128**, 010401 (2022).

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